

AN INTRODUCTION TO STATISTICS USING R



BY

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*Illustration from Alexander Pushkin's "Eugene Onegin."
The poem has been object of the first statistical study
involving Markov Chains.*

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1

PROBABILITY

1.1 INTRODUCTION

s:intro

Probability Theory is a mathematical framework introduced for quantifying uncertainty. Example of sentences with which it works are

e:stat1

1. The probability that tomorrow will rain is 0.75.
2. The probability that the coin that I am about to toss will show its head is $1/2$.

For the rest of the notes, we take a subjective stance and consider the above sentences as expressing a *subjective* degree of belief that an individual has for the events.

Many ways have been tried to give a rigorous meaning of probabilistic statements as the above ones in many different context. Those interesting discussions, however, are not needed when studying probability theory. Indeed, probability theory gives a formal framework to work with sentences such as the above ones, without the need for them to have a precise meaning nor to be true. For instance, statement 2 is not universally true since one can easily cheat with coin tosses and make heads always appear. This is not peculiar to probability theory alone: geometry works with points without never defining them and logic is concerned on how to work with sentences without never being concerned on the veracity of them.

To get an idea of which objects we need to formalise, we now proceed to decompose and to analyse the above statements. Both of them are of the form "The probability of *something* is a number". That *something* in the first case is $E = \text{"Tomorrow will rain"}$, and in the second case $F = \text{"The coin that I am about to toss will show heads"}$, and it will be called an *event*. Events are the objects to which probability refers and the probability of an event is a number, 0.75 in the first statement and 0.5, in the second. We now make an intermediate step and rewrite the above statements as

- The probability of E is 0.75
- The probability of F is 0.5

Therefore, the probability of an event is a number and the probability is an operation that associates to each event a number. Denoting "The probability of" by $\mathbb{P}()$, the above statements can be rewritten as

- $\mathbb{P}(E) = 0.75$
- $\mathbb{P}(F) = 0.5$.

and $\mathbb{P}$ can be seen as a function that takes in input an event and whose output is a number p between 0 and 1

$$\begin{array}{ccc} \mathbb{P} : \text{Events} & \mapsto & [0, 1] \\ E & \rightarrow & \mathbb{P}(E) \in [0, 1]. \end{array}$$

Therefore a typical probabilistic statement assumes the form of

- The probability of the event E is p , or, $\mathbb{P}(E) = p$,

where $p \in [0, 1]$.

1.2 EVENTS AND SAMPLE SPACE

Motivated by the Introduction 1.1 we first introduce the notion of events. We then see that events can be modelled using set theory, after introducing the sample space

1.2.1 Definition of an event

In light of the subjective perspective on Probability Theory we will refer to the person who is evaluating the probabilities of events using "you". Following [definetti] we can define events as

d:event

Definition 1.1. An event is an unambiguous statement which can be either true or false and about which you are uncertain.

Informally, unambiguous statement means that a possible bet or insurance based upon it can be decided without question. Examples of events are: "The next ChatGPT sentence will be incorrect" or "Napoleon was born on March" (to decide this last statement, I should open Wikipedia). Events will be usually denoted by capital letters, $A, B, \dots$, and the impossible event will be usually denoted by $\emptyset$.

1.2.2 Operations and relations between events

ss:operations

From two events E and F it is possible to build new events using the operations "and", "or" and "not", namely, to consider " E and F ", " E or F " and "not E ". We will write them as

$$E \wedge F = \text{"E and F"}, \quad E \vee F = \text{"E or F"}, \quad \neg E = \text{"not E"}.$$

e:and'or'not

The event $E \wedge F$ is true when both E and F are true, $E \vee F$ is true when at least one of the two is true (possibly both), and $\neg E$ is true when E is false. To make a simple example, let $E = \text{"Tomorrow it will rain"}$ and $F = \text{"The coin I am about to toss will show heads"}$, the two events of the Introduction. Then

- $E \wedge F = \text{"Tomorrow it will rain and the coin will show heads"}$.
- $E \vee F = \text{"Either tomorrow it will rain or the coin will show heads (or both)"}$.
- $\neg E = \text{"Tomorrow it will not rain"}$, $\neg F = \text{"The coin will not show heads"}$.

Given two events E and F we say that E *implies* F , and we write $E \rightarrow F$, if F is true whenever E is. If $G = \text{"Tomorrow it will rain heavily"}$, then $G \rightarrow E$. Note that in general

- $E \wedge F \rightarrow E$ and $E \wedge F \rightarrow F$;
- $E \rightarrow E \vee F$ and $F \rightarrow E \vee F$;

- if E is impossible, then $E \rightarrow F$ for every event F , and if F is certain, then $E \rightarrow F$ for every event E .

The first two relations say that "and" makes events smaller (harder to be true), while "or" makes them bigger.

^a

^a Different sentences can define the same event: $E =$ "The die shows an even number" and $F =$ "The die shows 2, 4 or 6" are the same event. We identify those two propositions with the same event and we write $E = F$. In particular, any two impossible events are equal and we denote them $\emptyset$.

1.2.3 Algebra of events

There are no inherent restrictions on the events one can define and on which you could try to assign a probability. When doing a probabilistic study, however, you fix beforehand the events you want to consider, and this choice depends on what you are looking at. If we are interested in the result of a coin toss, we want to consider the event "The coin shows heads", but not the event "The coin shows heads and tomorrow it will rain in Rome".

Example 1.1: The shop

ex:shop

You are paid by a shop owner to perform a statistical analysis of the customers visiting his shop. For time and economic constraints, you have to choose what kind of events you are interested in. One possibility is to consider only the number of customers, so that the events will be of the form "More than 10 customers will visit the shop", "3 customers will visit the shop", "Between 5 and 7 customers will visit the shop", ... If the shop owner is also interested in the gender of the customers, you can do a finer analysis and consider events that take into account both the number of customers and their gender, such as "More than 3 male customers and less than 10 female customers will visit the shop" or "2 male and 1 female customers will visit the shop".

The events we want to consider are called *measurable events*, since they are the ones that our observation is able to decide (in Example 1.1, by counting the customers) and, as we will see, the ones to which we assign a probability. Their set will be denoted by $\mathfrak{F}$.

The choice of $\mathfrak{F}$ is free, up to one requirement: if we want to consider E and F , we also have to consider $E \wedge F$, $E \vee F$ and $\neg E$. One of the reasons for this is that whatever we are looking at in order to decide whether E and F are true decides also whether $E \wedge F$, $E \vee F$ and $\neg E$ are true. In Example 1.1, if by counting the customers we can decide $E =$ "More than 10 customers will visit the shop" and $F =$ "Less than 20 customers will visit the shop", by counting the customers we can also decide $E \wedge F =$ "Between 11 and 19 customers will visit the shop" and $\neg E =$ "At most 10 customers will visit the shop".

We observe that when we will be introducing a probability in Section 1.4, we need to introduce it in such a way that the probabilities of $E \wedge F$, $E \vee F$ and $\neg E$ are tied to the ones of E and F . For instance, the probability of $\neg E$ is one minus the probability of E : assigning a probability to E means assigning a probability to $\neg E$ as well, whether we want it or not. Finally, we ask that

the certain event belongs to $\mathfrak{F}$: it is decided by any observation whatsoever, and it is the event to which we will give probability 1 and with which all the other probabilities are compared.

d:measurable

Definition 1.2 (Measurable events). The events we want to consider are called *measurable events*. Their set $\mathfrak{F}$ is an *algebra of events*, that is,

1. the certain event Ω belongs to $\mathfrak{F}$;
2. if $E \in \mathfrak{F}$, then $\neg E \in \mathfrak{F}$;
3. if $E, F \in \mathfrak{F}$, then $E \wedge F \in \mathfrak{F}$ and $E \vee F \in \mathfrak{F}$.^b

^b If we also ask that $E_1 \vee E_2 \vee E_3 \vee \dots$ belongs to $\mathfrak{F}$ whenever $E_1, E_2, E_3, \dots$ do, $\mathfrak{F}$ is called a σ -algebra of events. This is the notion used in measure theory.

Example 1.2: The algebra of a single event

ex:cointoss

Suppose that we want to consider only one event E . By Definition 1.2, $\mathfrak{F}$ has to contain also $\neg E$, Ω and $\emptyset$, and

$$\mathfrak{F} = \{\emptyset, E, \neg E, \Omega\} \quad (1)$$

e:cointoss

is an algebra of events: $E \wedge \neg E = \emptyset$, $E \vee \neg E = \Omega$, $\neg(\neg E) = E$, and "and", "or" and "not" of events of $\mathfrak{F}$ give again events of $\mathfrak{F}$, as you can check with the rules (??). It is the smallest algebra containing E . For instance, if we toss a coin and we are interested only in the face it shows, then E = "The coin shows heads", $\neg E$ = "The coin shows tails", Ω = "The coin shows either heads or tails", and (1) is the algebra of the coin toss.

Example 1.3: The shop, continued

ex:shop'algebra

In the first choice of Example 1.1, the measurable events are all the statements about the number of customers, such as "More than 10 customers will visit the shop", "Between 5 and 7 customers will visit the shop" or "An even number of customers will visit the shop", and their set is an algebra of events: "and", "or" and "not" of statements about the number of customers are again statements about the number of customers. We denote this algebra by $\mathfrak{F}_1$. In the second choice, the measurable events are all the statements about the number of male customers and the number of female customers, and their set $\mathfrak{F}_2$ is again an algebra of events.

1.2.4 Elementary events and sample space

ss:elementary

Once the events we want to consider are fixed and collected in $\mathfrak{F}$, it is possible to define a particular kind of events, the *elementary events*, thanks to which we can model events using set theory. Informally, elementary events

are the smallest among the events we want to consider, and can be regarded as the outcomes of the “experiment” we are observing. It is indeed possible to define a notion of smallness using the relation “implies” introduced in Section 1.2.2: an event E is smaller than F if $E \rightarrow F$, since E is true in fewer circumstances than F . The impossible event is the smallest of all, since $\emptyset \rightarrow F$ for every F , and the elementary events are the smallest ones after it.

d:elementary

Definition 1.3 (Elementary events). An event $E \in \mathfrak{F}$ is *elementary* if it is not impossible and the only events of $\mathfrak{F}$ that imply it are $\emptyset$ and E itself: if $F \in \mathfrak{F}$ and $F \rightarrow E$, then $F = \emptyset$ or $F = E$.

For instance, if $\mathfrak{F}$ consists of all the statements about the result of a die, “The result is 6” is elementary, while “The result is an even number” is not, since “The result is 2” implies it.

Ex. 1.2.1 — Show that two different elementary events are mutually exclusive, that is, that if E and F are elementary and $E \neq F$, then $E \wedge F = \emptyset$.

Answer (Ex. 1.2.1) — By Definition 1.2, $E \wedge F \in \mathfrak{F}$. Since $E \wedge F \rightarrow E$ and E is elementary, either $E \wedge F = \emptyset$ or $E \wedge F = E$. In the second case $E \rightarrow F$ and, since F is elementary and $E \neq \emptyset$, $E = F$, which is excluded. Hence $E \wedge F = \emptyset$.

Ex. 1.2.2 — In the algebra $\mathfrak{F}_1$ of Example 1.3, which of the following events are elementary? “No customer will visit the shop”, “At most one customer will visit the shop”, “Exactly 3 customers will visit the shop”.

Answer (Ex. 1.2.2) — “No customer will visit the shop” is E_0 and “Exactly 3 customers will visit the shop” is E_3 , and both are elementary: a statement about the number of customers that implies them is either the same statement or impossible. “At most one customer will visit the shop” is $E_0 \vee E_1$, and it is not elementary, since E_0 implies it and E_0 is neither impossible nor equal to it.

The notion of elementary event is fundamental since it allows to encode the events of $\mathfrak{F}$ as sets, see Section 1.2.5.

d:sample space

Definition 1.4 (Sample space). Given $\mathfrak{F}$, the *sample space* is the set of the elementary events of $\mathfrak{F}$. We denote it by Ω , the same symbol used for the certain event.^c

Elements of the sample space, that is, elementary events, are usually denoted by ω . The sample space can be regarded as the set of the possible outcomes of the system you are looking at, with the degree of precision that you want to impose. From now on we assume that every event of $\mathfrak{F}$ which is not impossible is implied by at least one elementary event.^d Some examples are in order.

Example 1.4: Sample space of a coin toss

If $\mathfrak{F}$ is the algebra of the coin toss (1), the elementary events are $\omega_1 = E$ = “The coin shows heads” and $\omega_2 = \neg E$ = “The coin shows tails”, and the sample space is $\Omega = \{\omega_1, \omega_2\}$. Note that we can use a differ-

^c The two uses of Ω are reconciled in Section 1.2.5, where the certain event is identified with the set of all the elementary events.

^d This is always the case when $\mathfrak{F}$ is finite, and it is the case in all the examples of these notes.

There are algebras of events with no elementary event at all: the statements about an infinite sequence of coin tosses that involve only finitely many tosses, see Example 1.4, form one, since a condition on one more toss makes any such statement strictly smaller.

ent notation and set $1 = \omega_1$ and $0 = \omega_2$. However, 0 and 1 must not be mistaken for numbers: they are symbols indicating elementary events.^e

Example 1.5: A die is rolled

ex:die'sample

If a die is rolled and $\mathfrak{F}$ consists of all the statements about the result, such as "The result is an even number", the elementary events are the ones that cannot be decomposed just by looking at the die, $1 =$ "The result is 1", $\dots$, $6 =$ "The result is 6". The sample space is therefore $\Omega = \{1, 2, 3, 4, 5, 6\}$ and, again, the elements $1, \dots, 6$ must not be mistaken for numbers, since they are elementary events and, for instance, $1 + 3$ doesn't make sense.

Example 1.6: The shop, continued

ex:shop'sample

In the algebra $\mathfrak{F}_1$ of Example 1.3 the elementary events are $E_n =$ "Exactly n customers will visit the shop", for $n = 0, 1, 2, \dots$: a statement about the number of customers that implies E_n is either E_n itself or impossible. The sample space $\Omega = \{E_0, E_1, E_2, \dots\}$, which we can also write as $\{0, 1, 2, \dots\}$, has infinitely many elements. In the algebra $\mathfrak{F}_2$ the elementary events are $E_{m,f} =$ "Exactly m male and f female customers will visit the shop", and the sample space is the set of the pairs (m, f) .

Example 1.7: A chess match

If you observe a chess match, the events you want to consider are those that involve the game alone, such as "The game will end before 10 moves are taken" or "Black wins", while the elementary events are the possible single games. Therefore, the sample space is the set of all possible games. A full description of a game is usually given in terms of the so called algebraic notation; for instance, the fifth game of the 1972 World Championship between Spassky and Fischer is

1.d4 Nf6 2.c4 e6 3.Nc3 Bb4 4.Nf3 c5 5.e3 Nc6 6.Bd3 Bxc3+ 7.bxc3 d6 8.e4 e5 9.d5 Ne7 10.Nh4 h6 11.f4 Ng6 12.Nxg6 fxg6 13.fxe5 dxe5 14.Be3 b6 15.O-O O-O 16.a4 a5 17.Rb1 Bd7 18.Rb2 Rb8 19.Rbf2 Qe7 20.Bc2 g5 21.Bd2 Qe8 22.Be1 Qg6 23.Qd3 Nh5 24.Rxf8+ Rxf8 25.Rxf8+ Kxf8 26.Bd1 Nf4 27.Qc2 Bxa4 0-1

The algebraic notation can be seen as a way to encode elementary

events and the sample space. In Exercises 1.2.12 and 1.2.10 examples of sample spaces defined by simple games are given.

Ex. 1.2.3 — From the 2019-2020 course "Mathematics and Statistics" given by professors Khovanskaya, Bubilin, Shurov, Filimonov and Sonin at HSE. A six faced die is rolled. Find the elementary events that imply the event

1. "The result is 6"
2. "The result is a number less or equal 2"
3. "The result is an even number"
4. "The result is a number strictly greater than 4"
5. "The result is seven"

Answer (Ex. 1.2.3) — 1. Only "The result is 6" itself.

2. "The result is 1" and "The result is 2", since "The result is a number less than or equal to 2" = "The result is 1" $\vee$ "The result is 2".
3. "The result is 2", "The result is 4" and "The result is 6".
4. "The result is 5" and "The result is 6".
5. None. The event is impossible, that is, it is $\emptyset$, and elementary events are not impossible.
6. It is already an elementary event
7. "The result is a number less or equal than 2" = "The result is 1" or "The result is 2".
8. "The result is a number strictly greater than 4" = "The result is 5" or "The result is 6".
9. The event is the impossible event, also denoted by $\emptyset$.

Ex. 1.2.4 — A coin is tossed twice. We are interested in the face which the coin shows when it lands: heads or tails. The set of elementary events is $\{HH, HT, TH, TT\}$, where the event HH corresponds to "the first toss gave heads, the second heads", the event HT corresponds... Write the elementary events that imply the events

1. "We obtained two heads".
2. "The first toss gave heads".
3. "We obtained 1 tails "
4. "At least one toss gave tails"

Answer (Ex. 1.2.4) — 1. HH.

2. HH and HT.
3. HT and TH.
4. HT, TH and TT.

Ex. 1.2.5 — A coin is tossed 4 times. We are interested in the face that it shows when it lands: heads or tails. How many elementary events are there? which elementary events are contained in the events

1. The first result was heads
2. The second result was tails
3. The first result was heads and the second tails
4. All the 4 tosses gave the same result

Answer (Ex. 1.2.5) — The elementary events are the strings of four letters H or T , $HHHH, HHHT, HHTH, \dots, TTTT$, and there are $2^4 = 16$ of them.

1. The eight strings starting with H : $HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT$.
2. The eight strings with T in the second position: $HTHH, HTHT, HTTH, HTTT, TTHH, TTHT, TTTH, TTTT$.
3. The four strings starting with HT : $HTHH, HTHT, HTTH, HTTT$.
4. $HHHH$ and $TTTT$.

ercise:marbles

Ex. 1.2.6 — ([Ross] Chapter 2 Exercise 1) A box contains 3 marbles: 1 red, 1 green, 1 blue. Consider the experiment that consists in taking 1 marble from the box and then replacing it in the box and then drawing a second marble. Describe the sample space.

Answer (Ex. 1.2.6) — The sample space is the set of the outcomes of the experiment, "The first marble is red and the second is red", "The first marble is red and the second is green", and so on. A shorter way to write an outcome is a pair (r, g) , where the first entry is the colour of the first marble and the second entry the colour of the second one. The sample space is then

$$\Omega = \{(r, r), (r, g), (r, b), (g, r), (g, g), (g, b), (b, r), (b, g), (b, b)\}.$$

e:marbles

Ex. 1.2.7 — Repeat Exercise 1.2.6, when the second marble is drawn without replacing the first marble.

Answer (Ex. 1.2.7) — The outcomes (r, r) , (g, g) and (b, b) are no longer possible, and the sample space is

$$\Omega = \{(r, g), (r, b), (g, r), (g, b), (b, r), (b, g)\}.$$

e:marbles2

Ex. 1.2.8 — You toss a coin 3 times. Describe the elementary events.

Answer (Ex. 1.2.8) — $\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$, where, for instance, HTH is the event "The first toss gave heads, the second tails and the third heads".

1.2.5 Events as subsets of the sample space

ss:subset
d:favour

Definition 1.5 (Elementary events in favour of an event). We say that an elementary event $\omega \in \Omega$ is *in favour* of $E \in \mathfrak{F}$ if $\omega \rightarrow E$ (ω implies E).

Remark 1.6. The minimality of the elementary events implies that for each $E \in \mathfrak{F}$ and $\omega \in \Omega$ either ω is in favour of E or the two are mutually exclusive, $\omega \wedge E = \emptyset$: indeed $\omega \wedge E \rightarrow \omega$, so that $\omega \wedge E = \emptyset$ or $\omega \wedge E = \omega$, and in the second case $\omega \rightarrow E$.

We can associate to each event the subset of Ω whose elements are the elementary events in its favour:

$$\begin{aligned} \Phi : \mathfrak{F} &\rightarrow \{A, A \subset \Omega\} \\ E &\mapsto \{\omega \in \Omega, \omega \text{ is in favour of } E\}. \end{aligned}$$

e:abstract to c

Example 1.8: Die

ex:die'subset

We have seen in Example 1.5 that the sample space of a rolled die can be identified with $\Omega = \{1, 2, 3, 4, 5, 6\}$. The elementary events in favour of $E = \text{"The result is an odd number"}$ are 1, 3, 5. Therefore we identify E with the subset $\Phi(E) = \{1, 3, 5\}$ of Ω .

The fundamental fact about Φ is that the operations "and", "or" and "not" correspond to the operations $\cap$, $\cup$ and $()^c$ between subsets of Ω , that "implies" corresponds to "is a subset of", and that different events correspond to different subsets.

p:subset

Proposition 1.7. Let $E, F \in \mathfrak{F}$. Then

1. $\Phi(E \wedge F) = \Phi(E) \cap \Phi(F)$, $\Phi(E \vee F) = \Phi(E) \cup \Phi(F)$ and $\Phi(\neg E) = \Phi(E)^c$, the set of the elementary events that are not in $\Phi(E)$;
2. $E \rightarrow F$ if and only if $\Phi(E) \subset \Phi(F)$;
3. $E = F$ if and only if $\Phi(E) = \Phi(F)$.

Moreover, $\Phi(\emptyset)$ is the empty set and $\Phi(\Omega) = \Omega$, the whole sample space.

Proof. Since $E \wedge F$ is true exactly when E and F are both true, $\omega \rightarrow E \wedge F$ if and only if $\omega \rightarrow E$ and $\omega \rightarrow F$, which is the first identity. For the third one, if $\omega \rightarrow E$ and $\omega \rightarrow \neg E$, then $\omega \rightarrow E \wedge \neg E = \emptyset$, and ω would be impossible: hence $\omega \in \Phi(\neg E)$ implies $\omega \notin \Phi(E)$. Conversely, if $\omega \notin \Phi(E)$, then $\omega \wedge E = \emptyset$ and, by the rules (??), $\omega = \omega \wedge (E \vee \neg E) = (\omega \wedge E) \vee (\omega \wedge \neg E) = \omega \wedge \neg E$, that is, $\omega \rightarrow \neg E$. The second identity follows from the other two, since $E \vee F = \neg(\neg E \wedge \neg F)$ by de Morgan's laws. As for point 3, suppose that $\Phi(E) = \Phi(F)$ but $E \neq F$. Then one between $E \wedge \neg F$ and $F \wedge \neg E$ is not impossible, since $E \wedge \neg F = \emptyset$ gives $E = (E \wedge F) \vee (E \wedge \neg F) = E \wedge F$, that is, $E \rightarrow F$, and in the same way $F \wedge \neg E = \emptyset$ gives $F \rightarrow E$. Say that $E \wedge \neg F \neq \emptyset$. By the assumption made after Definition 1.4, there is an elementary event $\omega \rightarrow E \wedge \neg F$, so that $\omega \in \Phi(E)$ and $\omega \notin \Phi(F)$, which contradicts $\Phi(E) =$

$\Phi(F)$. Point 2 follows: $E \rightarrow F$ if and only if $E \cap F = E$ (see the exercises of Section 1.2.3), if and only if $\Phi(E) \cap \Phi(F) = \Phi(E)$ by points 1 and 3, if and only if $\Phi(E) \subset \Phi(F)$. Finally, no elementary event is in favour of $\emptyset$, since elementary events are not impossible, and every elementary event is in favour of Ω . $\square$

From now on we will consider $\mathfrak{F}$ as a set of subsets of Ω , that is, we will not distinguish between an event E and the set $\Phi(E)$ of the elementary events in its favour. This allows us to depict events, their unions and their intersections with Venn's diagrams, as Figure 1.1 shows.

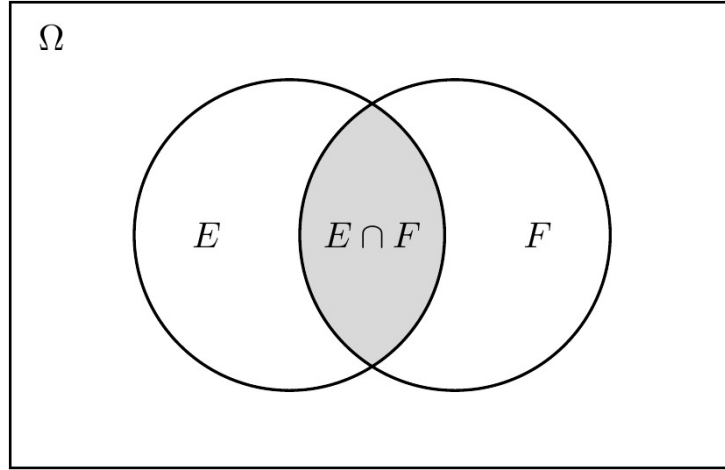


Figure 1.1: Venn's diagram

f:Venn

Ex. 1.2.9 — ([Feller], Chapter 1.8) Let Ω be a sample space and A, B, C be three arbitrary events. Find the expressions for the events that of A, B, C :

1. Only A occurs.
2. Both A and B , but not C occur.
3. All three events occur.
4. At least one occurs.
5. At least two occur.
6. One and no more occurs.
7. Two and no more occur
8. None occurs.
9. Not more than two occur.

Answer (Ex. 1.2.9) — 1. $A \cap B^c \cap C^c$.

2. $A \cap B \cap C^c$.

3. $A \cap B \cap C$.

4. $A \cup B \cup C$.

5. At least two occur if one between $A \cap B$, $A \cap C$ and $B \cap C$ occurs, that is, $D = (A \cap B) \cup (A \cap C) \cup (B \cap C)$.

6. One of the three occurs, that is, $A \cup B \cup C$ occurs, and two of them do not occur, that is, D does not occur: $(A \cup B \cup C) \cap D^c$.

7. Two of them occur, that is, D occurs, and all three do not: $D \cap (A \cap B \cap C)^c = D \cap (A^c \cup B^c \cup C^c)$.
8. $(A \cup B \cup C)^c = A^c \cap B^c \cap C^c$.
9. This is the event "all three occur" not occurring: $(A \cap B \cap C)^c = A^c \cup B^c \cup C^c$.

1.2.6 Relevant Examples

ss:models

In this section we collect the basic sample spaces to which we will come back throughout the notes: the sample space of the *Bernoulli trials* and the one of a random number.

ex:one'coin

Model 1.1 (A coin toss). If in a coin toss we are interested only in the face the coin shows, as we always are when we speak about coin tosses, then the only two non-trivial events we want to consider are 0 = "The result is tails" and 1 = "The result is heads", see Examples 1.2 and 1.4. Therefore

$$\Omega_1 = \{0, 1\},$$

and we will often refer to 1 as a success. The set of events we want to consider is

$$\mathfrak{F} = \{\emptyset, \{0\}, \{1\}, \{0, 1\}\}.$$

ex:n'coin

Model 1.2 (Two, three and n coin tosses). If we toss a coin twice and we are only interested in the faces it shows, the events we want to consider are of the form "The first toss gave heads" or "The second toss gave tails", and the elementary events are of the form "The first toss gave tails and the second heads", which we encode in the two-digit string 01 . Therefore the sample space is

$$\Omega_2 = \{00, 01, 10, 11\} = \{x_1 x_2, x_i \in \{0, 1\} \text{ for } i = 1, 2\},$$

the set of the two-digit strings where every digit represents the outcome of the corresponding toss. Similarly, for three coin tosses

$$\Omega_3 = \{000, 001, 010, 011, 100, 101, 110, 111\} = \{x_1 x_2 x_3, x_i \in \{0, 1\} \text{ for } i = 1, 2, 3\},$$

and, in the general case of n coin tosses, the sample space is the set of the strings of n digits with values in $\{0, 1\}$,

$$\Omega_n = \{x_1 x_2 \dots x_n, x_i \in \{0, 1\} \text{ for } i = 1, \dots, n\}. \quad (2)$$

e:ntoss

Events that we will often use, when introducing the binomial distribution, are E_k = "There have been exactly k successes", for $k = 0, 1, \dots, n$. For instance, if $n = 3$ and $k = 1$, then $E_1 = \{001, 010, 100\} = \{x_1 x_2 x_3, x_1 + x_2 + x_3 = 1\}$. The encoding (2) allows us to write the condition imposed by E_k in simple terms:

$$E_k = \{x_1 \dots x_n \in \Omega_n, x_1 + \dots + x_n = k\}.$$

e:ntoss'ksucce

ex:finite'coin

Model 1.3 (A finite but undetermined number of coin tosses). If we observe a finite number of coin tosses, but we do not know in advance how many of them there will be, the elementary events are the strings of any length,

$$\begin{aligned}\Omega &= \Omega_1 \cup \Omega_2 \cup \Omega_3 \cup \dots = \{0, 1, 00, 01, 10, 11, 000, \dots\} \\ &= \{x_1 \dots x_n, n \in \mathbb{N}, x_i \in \{0, 1\} \text{ for } i = 1, \dots, n\},\end{aligned}$$

e:finite'coin

where Ω_n has been defined in (2).

r:uncountable

Remark 1.8. The sample spaces above have the important property that their elements can be written as a list, $\Omega = \{\omega_1, \dots, \omega_n\}$ for some $n \in \mathbb{N}$ in the finite case, or $\Omega = \{\omega_1, \omega_2, \dots\}$ in the infinite case. In other words, we have found a rule to enumerate the elements of Ω , and every element of Ω is reached by this rule. This is no longer possible in the next two models (try yourself to enumerate the elements of Ω). In mathematical terms, in the next two models Ω is uncountably infinite.^f

^e Despite that almost every textbook in probability starts by providing the sample space, the sample space in probability is only used as a mean of ground events to a standard object, see Subsection 1.2.5

x:infinite'coin

Model 1.4 (Infinite coin tosses). If we observe an infinite sequence of coin tosses, the events we want to consider are the statements that involve finitely many tosses, such as

$$A_{a_1 \dots a_n} = \text{"The first toss gave } a_1, \text{ the second } a_2, \dots, \text{ the } n\text{-th } a_n\text{"}, \quad (3)$$

e:cylinder

where $n \in \mathbb{N}$ and $a_i \in \{0, 1\}$ for $i = 1, \dots, n$, together with their "and", "or" and "not". The sample space is the space of infinite strings

$$\Omega = \{x_1 x_2 \dots x_n \dots, x_i \in \{0, 1\}\},$$

e:infinite'coin

and we identify the event (3) with the set of the strings whose first n digits are $a_1, \dots, a_n$,

$$A_{a_1 \dots a_n} = \{x_1 x_2 \dots \in \Omega, x_1 = a_1, \dots, x_n = a_n\}.$$

e:infinite'coin'set

8

ex:unif

Model 1.5 (A random number between 0 and 1). By typing on R the command `runif`, the computer gives us a number X .

^f For mathematical reasons, the algebra of event $\mathfrak{F}$ most of the times does not contain every subset of Ω

```
runif(1)
```

```
## [1] 0.3612239
```

The events we are interested in are of the form " $X < 1/2$ " or " $X \in [1/4, 5)$ ", that is, " $X \in I$ " for an interval $I \subset \mathbb{R}$, together with their "and", "or" and "not". As in Model 1.4. The number π

$$\Omega = [0, 1]$$

e:unif 1

as the set of the outcomes, and to identify the event " $X \in I$ " with the interval I . We wi

1.2.7 Exercises

exercise:game'2

Ex. 1.2.10 — ([Ross] Chapter 2 Exercise 2) In an experiment, die is rolled continually until a 6 appears, at which point the experiment stops. What is the sample space of this experiment? Let E_n denote the event "The experiment is completed before n rolls". Write it as a subset of the sample space. Describe $(\bigcup_{i=1}^{\infty} E_i)^c$.

Answer (Ex. 1.2.10) — We encode the outcome "The first roll gave 4, the second 2, the third 6, and then the experiment stopped" in the string 426: a digit is the result of a roll, its position is the number of the roll, and the string stops after the first 6. We also have to allow for the die never showing a 6, which corresponds to the infinite strings of digits between 1 and 5, such as 121212... Therefore

$$\Omega = \{6, 16, 26, 36, 46, 56, 116, \dots\} \cup \{x_1 x_2 \dots x_n \dots, x_i \in \{1, 2, 3, 4, 5\}\} = E \cup G.$$

The set E is the event "The experiment stops after finitely many rolls" and G is the event "The experiment never stops". As in Model 1.4, the strings of G are outcomes rather than events, since none of them can be decided with finitely many rolls; for any reasonable probability G has probability 0, but for other games one should take into account the event that the game never ends and make sure it has probability 0. The events E_n are $E_1 = \{6\}$, $E_2 = \{6, 16, 26, \dots, 56\}$, $E_3 = E_2 \cup \{116, 126, \dots, 556\}$, and in general

$$E_n = \{x_1 \dots x_m 6, x_j \in \{1, \dots, 5\}, m \leq n-1\},$$

the strings of length at most n . By definition of "or", $E_1 \cup E_2 \cup E_3 \cup \dots$ is the event " E_n occurs for at least one n ", that is, "The experiment stops after finitely many rolls", which is E . Its complement is G .

The event

$$\bigcup_{i=1}^{\infty} E_n = E_0 \cup E_1 \cup E_2, \dots = "E_0 \text{ or } E_1 \text{ or } \dots E_n \text{ or } \dots"$$

is, by definition of "or", the event that E_n happens for at least one n . That is, is the event "The game ends before n rolls at least one n " = "The game ends after n rolls for some n ", which coincides with E . The set event E^c corresponds to G .

Ex. 1.2.11 — ([Ross] Chapter 2 Exercise 3) Two dice are thrown. Let E be the event that the sum of the dice is odd, and F the event that at least one of the two dice lands in 1 and let G be the event that the sum is 5. Describe the events $E \cap F$, $E \cup F$, $F \cap G$, $E \cap F^c = E \setminus F$ and $E \cap F \cap G$

Answer (Ex. 1.2.11) — Answer (Ex. 1.2.11) — Writing ij for “The first die shows i and the second j ”,

$$E = \{12, 14, 16, 21, 23, 25, 32, \dots, 65\}, \quad F = \{11, 12, 13, 14, 15, 16, 21, 31, 41, 51, 61\}, \quad G = \{14, 23, 32, 41\}.$$

Then $E \cap F =$ “One die shows 1 and the other an even number” $= \{12, 14, 16, 21, 41, 61\}$;
 $E \cup F =$ “The sum is odd or a die shows 1” $= \{11, 12, 13, 14, 15, 16, 21, 23, 25, 31, 32, 34, \dots, 65\}$;
 $F \cap G =$ “One die shows 1 and the other 4” $= \{14, 41\}$; $E \cap F^c =$ “The sum is odd and no die shows 1” $= \{23, 25, 32, 34, 36, 43, 45, 52, 54, 56, 63, 65\}$; finally $G \subset E$, since a sum equal to 5 is odd, so that $E \cap F \cap G = F \cap G = \{14, 41\}$.

e:game

Ex. 1.2.12 — ([Ross] Chapter 2 Exercise 4) Anton, Barbara and Carlo take turns flipping a coin. The first one to get head wins. The sample space to this experiment can be defined as $\Omega = \{1, 01, 001, 0001, \dots\} \cup \{00000\dots\}$.

1. Interpret the sample space.
2. In terms of the sample space, write the following events: $A =$ “Anton wins”, $B =$ “Barbara wins” and $(A \cap B)^c$.

Answer (Ex. 1.2.12) — 1. 1 = “The first toss gave heads”, so that Anton wins; 01 = “The first toss gave tails and the second heads”, so that Barbara wins; 001 is the event that Carlo wins at the third toss, and so on. The string 000... is the event “Heads never appears”, so that the game never ends.

2. Anton tosses at the first, fourth, seventh, ... turn, so he wins when the 1 is in position $3k + 1$ for some $k \in \mathbb{N}$:

$$A = \{1, 0001, 0000001, \dots\}.$$

In the same way, Barbara wins when the 1 is in position $3k + 2$:

$$B = \{01, 00001, 00000001, \dots\}.$$

Finally $(A \cup B)^c = \{001, 000001, \dots\} \cup \{000\dots\}$ is the event “Carlo wins or the game never ends”.

1.3 PROBABILITY ON DISCRETE SAMPLE SPACES

1.4 PROBABILITY

1.4.1 Introduction

In this section we formally define what a probability is. As already said in the Introduction 1.1, a probability associates to each event $E \in \mathfrak{F}$, the set of events that we want to consider, its probability $\mathbb{P}(E)$. A probability is therefore a function

$$\begin{aligned} \mathbb{P} : \mathfrak{F} &\rightarrow [0, 1] \\ E &\mapsto \mathbb{P}(E). \end{aligned}$$

e:prob'def

Not every function from $\mathfrak{F}$ to $[0, 1]$ has the right to be called a probability, and some relations must be satisfied. Consider the two events $E = \text{"Tomorrow it will be cloudy"}$ and $F = \text{"Tomorrow it will rain"}$ and their probabilities $\mathbb{P}(E)$ and $\mathbb{P}(F)$. If F happens, also E happens, that is, $F \subset E$, and the intuition tells us that the numbers $\mathbb{P}(E)$ and $\mathbb{P}(F)$ have to satisfy $\mathbb{P}(F) \leq \mathbb{P}(E)$. This monotonicity property, for instance, will be proved in Corollary 1.17 as a consequence of the properties we impose on probabilities, see Definition 1.10.

The relations we require are two, see Definition 1.10: the certain event has probability 1, and the probability of the "or" of two mutually exclusive events is the sum of their probabilities. Both make sense on any algebra of events, see Definition 1.2, and they are weaker than the ones required in measure theory, where the second relation is asked for infinitely many events $E_1, E_2, E_3, \dots$ at once, so that $\mathfrak{F}$ has to be a σ -algebra of events.^h In these notes we mostly consider algebras $\mathfrak{F}$ whose sample space Ω is finite. In this case a probability is determined by the values $\mathbb{P}(\omega)$ it assigns to the elementary events $\omega \in \Omega$, see Theorem ??, and one can skip directly to Section ??.ⁱ

^h The second relation is called additivity, and its version for infinitely many events σ -additivity. Requiring additivity alone is also the position of [definetti].

ⁱ When Ω is infinite, as in Example 1.6, the values $\mathbb{P}(\omega)$ determine $\mathbb{P}$ only if σ -additivity is assumed.

For some of the sample spaces we are interested in, however, the difference between algebras and σ -algebras matters. In Exercise 1.2.10 the event $G = \text{"The game never ends"}$ is the "and" of the infinitely many events "The experiment is not completed before n rolls", $n \in \mathbb{N}$, that is, $G = \bigcap_{n=1}^{\infty} E_n^c$, and in Model 1.5 the event " $X = a$ " is the "and" of the events " $X \in [a - \frac{1}{n}, a + \frac{1}{n}]$ ", $n \in \mathbb{N}$. An algebra of events containing the E_n need not contain G , and computing the probability of G from the probabilities of the E_n requires σ -additivity. We will not treat these cases rigorously: we will only compute the probabilities of some particular events of Model 1.4, when dealing with Bernoulli trials, see Definition ??, and of Model 1.5, when dealing with continuous random variables, see Definition ??.

1.4.2 Introduction

In this section we are going to formally define what is a probability. As already said in the Introduction 1.1, a probability associates to each $E \in \mathfrak{F}$, the set of events that we want to consider, its probability, $\mathbb{P}(E)$. A probability is therefore a function

$$\begin{aligned} \mathbb{P} : \mathfrak{F} &\mapsto [0, 1] \\ E \in \mathfrak{F} &\rightarrow \mathbb{P}(E) \end{aligned}$$

e:prob'def

Not every function from $\mathfrak{F}$ to $[0,1]$ has the right to be called a probability, and some relations must be satisfied. Consider the example of the two events $E = \text{"Tomorrow will be cloudy"}$ and $F = \text{"Tomorrow it will rain"}$ and of its associated probabilities $\mathbb{P}(E)$ and $\mathbb{P}(F)$. If F happens, also E happens, and the intuition tells us that the numbers $\mathbb{P}(E)$ and $\mathbb{P}(F)$ have to verify $\mathbb{P}(F) \leq \mathbb{P}(E)$. This monotonicity property, for instance, will be proved in Corollary 1.17

In this notes we will not introduce the definition of probability in the full general context. We rather restrict to finite sample space $\Omega = \{\omega_1, \dots, \omega_n\}$, for some $n \in \mathbb{N}$. Up to replacing finite sums with infinite sums, all the definitions and the results below generalise immediately to the discrete infinite setting $\Omega = \{\omega_1, \dots, \omega_n, \dots\}$, where the sample space can be written as a list of elements. Unfortunately, many sample spaces that are used in practice, such as $\Omega = [0,1]$, see Example 1.5 or $\Omega = \{x_1 x_2 \dots, x_i \in \{0,1\} \text{ for } i \in \mathbb{N}\}$ defined in 1.4, are simply too big to be written as a list of elements. In such cases the theory gets more involved and invokes mathematical results from measure theory. The reason is that the usual requirements that a function $\mathbb{P} : \mathfrak{F} \rightarrow [0,1]$, the *Kolmogorov axioms*, are too strict to allow each $E \subset \Omega$ to be in $\mathfrak{F}$, and one has to restrict to a so-called σ -algebra of subsets of Ω , a notion that this notes carefully avoid to introduce. Nevertheless, in such cases, we will evaluate the probabilities of particular events when dealing with Bernoulli trials, see Definition ?? and with continuous random variables, see Definition ??.

1.4.3 Definition of a probability on a finite sample space

Hereafter we consider a finite sample space $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$, for some $n \in \mathbb{N}$, and we consider every event that can be written in terms of the elementary events $\omega_1, \dots, \omega_n$, namely

$$\mathfrak{F} = \{A, A \subset \Omega\}$$

e:fall

consists in every subset of Ω (including the impossible event $\emptyset$ and the certain event Ω). The particularity of the discrete case we are considering is that in order to define a probability on Ω it is sufficient to define the probabilities of the elementary events

d:prob`elementary

Definition 1.9 (Probability of the elementary events). A probability for the elementary events $\omega_i, i = 1, \dots, n$ is a collection numbers $p_1, \dots, p_n$ verifying that $p_i \in [0,1]$ for each $i = 1, \dots, n$ and

$$p_1 + p_2 + \dots + p_n = 1 \tag{4}$$

e:normalization

The condition (10) has the intuitive meaning that we regard the event in which none of the ω_i happens as impossible, see Proposition 1.13.

Example 1.9

ex:definition

The sample space of a rolled die is $\Omega = \{1, 2, 3, 4, 5, 6\}$. Definition 1.9 allows for different choices of probabilities on elementary events, but a typical choice is

$$p_1 = \dots = p_6 = 1/6,$$

e:die'uni

which corresponds to a perfectly balanced die or a die about which we do not have any information. A probability corresponding to an unfair die could look like

$$p_1 = p_2 = p_3 = 1/4, \quad p_4 = p_5 = p_6 = 1/12. \quad (5)$$

e:die'unfair

The numbers p_i , for $i = 1, \dots, n$, are the probability of the elementary event ω_i , namely $\mathbb{P}(\{\omega_i\}) := p_i$. We will sometimes, for instance in (6), use the different notation $\mathbb{P}(\{\omega\}) = p_\omega$, which avoids to use the index i with which the elements of Ω are ordered and that, therefore, does not depend on the particular way the elements are ordered. If $\omega = \omega_i$, then the two notations are related by $p_i \equiv p_{\omega_i}$.

Example 1.10

coin'toss'notation

This example illustrates the difference in the two possible notations we have introduced. Consider Ω_2 , the sample space of two coin tosses given in Example 1.2. Ordering the elements $\Omega = \{00, 01, 10, 11\}$ by $\omega_1 = 00$, $\omega_2 = 01$, $\omega_3 = 10$ and $\omega_4 = 11$, then a possible probability using the notation introduced in Definition 1.9 is given by

$$p_1 = 1/8, p_2 = 1/8, p_3 = 1/2, p_4 = 1/4$$

and the same probability, using the equivalent notation p_ω , is given by

$$p_{00} = 1/8, p_{01} = 1/8, p_{10} = 1/2, p_{11} = 1/4$$

Note that the possible ways of choosing p_i , for $i = 1, \dots, n$, are quite arbitrary and the only restrictions involved in the choice is that $p_i \in [0, 1]$ and (10), which basically means that there are $n - 1$ numbers (which have to be positive and whose sum has to be less than 1) to fix. However, in the discrete setting, this is all the freedom at our disposal, and once the probabilities of the elementary events are assigned, the probability of every event $E \in \mathfrak{F}$ is determined by the sum of the probabilities of the elementary events that compose it, according to the following definition

d:prob

Definition 1.10. Let $p_1, p_2, \dots, p_n, n \in \mathbb{N}$ be a probability for the elementary events, see Definition 1.9, and let $A \subset \Omega$ be an event. The probability of A is the sum of the probabilities of the elementary events that compose it

$$\mathbb{P}(A) = \sum_{\omega \in A} p_{\omega} \quad (6) \quad \text{e:prob}$$

Using a different notation, any subset A of Ω can be written as

$$A = \{\omega_{i_1}, \dots, \omega_{i_k}\}, \quad (7) \quad \text{e:notation}$$

for some $k \leq n$ and set of indexes $1 \leq i_1 < i_2 < \dots < i_k \leq n$. Its probability is then given by

$$\mathbb{P}(A) = p_{\omega_{i_1}} + \dots + p_{\omega_{i_k}} \equiv p_{i_1} + \dots + p_{i_k} \quad (8) \quad \text{e:prob2}$$

Example 1.11

ex:dice`prob

We come back to Example 1.9 in order to see Definition 1.10 in action and to get better acquainted with the two different notations (6) and (8). The event "The result is an odd number" corresponds to the subset $\{1, 3, 5\}$ and (6) tells that the two probabilities $\mathbb{P}$ and $\mathbb{Q}$ defined by the choices (??) and (5), respectively, evaluate the probability of this event by

$$\mathbb{P}(\{1, 3, 5\}) = \mathbb{P}(\{1\}) + \mathbb{P}(\{3\}) + \mathbb{P}(\{5\}) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

and

$$\mathbb{Q}(\{1, 3, 5\}) = \mathbb{Q}(\{1\}) + \mathbb{Q}(\{3\}) + \mathbb{Q}(\{5\}) = \frac{1}{4} + \frac{1}{4} + \frac{1}{12} = \frac{7}{12},$$

respectively. In order to use the notation in (8), we rewrite the set $\{1, 3, 5\}$ as in (7) by setting $k = 3$ and $i_1 = 1, i_2 = 3$ and $i_3 = 5$. Observe also that $\mathbb{P}(\Omega) = \mathbb{Q}(\Omega) = 1$ thanks to (10).

e:die`prob`uni

e:die`prob`unfair

We finally collect the definitions of the present section

d:prob3

Definition 1.11. A *probability* over Ω is a function

$$\begin{aligned} \mathbb{P} : \mathfrak{F} &\rightarrow [0, 1] \\ E &\mapsto \mathbb{P}(E) \end{aligned}$$

such that $p_i := \mathbb{P}(\{\omega_i\})$ forms a probability on elementary events in the sense of Definition 1.9 and (6) (or (8)) holds.

1.4.4 Uniform probability

In the case in which you evaluate each elementary event equally likely to occur, or in the case in which you don't have any knowledge to say that

any particular elementary event is more likely to occur, a possible choice is to assign to each elementary event the same probability $p \in [0, 1]$. To determine p we use (10)

$$p_1 + \dots + p_n = \overbrace{p + \dots + p}^{n \text{ times}} = np = 1, \quad \text{e:uniform1}$$

from which we obtain that $p = 1/n$. To compute the probability of an event composed by k elementary events $A = \{\omega_{i_1}, \dots, \omega_{i_k}\}$, with $k \leq n$ and $1 \leq i_1 < \dots < i_k \leq n$ we use (8) to obtain

$$\mathbb{P}(A) = p_{i_1} + \dots + p_{i_k} = \overbrace{\frac{1}{n} + \dots + \frac{1}{n}}^{k \text{ times}} = \frac{k}{n}, \quad (9) \quad \text{e:uniform2}$$

which depends only on k , the number of elementary events that compose A . Denoting by $|E|$ the number of elements of the set $E \subset \Omega$, one can rewrite (9) as the number of elements that compose A divided by the total number of events

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

To get acquainted with the notation independent from the ordering used in (6), we repeat the same computations

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\}) = \sum_{\omega \in A} p = p \left(\sum_{\omega \in A} 1 \right) = p|A| = \frac{|A|}{|\Omega|},$$

and obtain

d:uniform

Definition 1.12 (Uniform Probability). Let Ω be a finite sample space. The uniform probability is defined by

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|}, \quad \text{e:uniform}$$

or each $E \subset \Omega$.

Basically, computing the uniform probability reduces to counting configurations. Unfortunately, despite being conceptually simple, it is often computationally unfeasible. The same holds for general discrete probabilities, which can be seen as a way to count objects giving different weights to them. The uniform probability is often mistakenly taken as the definition of probability. This has led to some abuse of notation such as the following

Example 1.12: Extracting Randomly a Person

ex:person

Consider extracting a person between 20, and denote $\Omega = \{1, \dots, 20\}$ the sample space, where i denotes the elementary event "the i -th person is extracted". Usually what is meant by the sentence "a person is extracted randomly" is that a person is selected with uniform probability, namely $\mathbb{P}(\{i\}) = 1/20$. However, from our point of view, a

Nobel prize is also chosen randomly among the possible candidates, but there is no reason why the probability should be the uniform one. Another that can be ambiguous are "we choose a person as randomly as possible", by meaning as fairly or as uniform as possible.

As a side note, despite being natural to model the choice of a person with uniform probability, when doing a survey or making a poll statisticians struggle to make their choice adhere to this model. When mobile phones were not so widespread, doing surveys by phone call excluded a portion of the population, for instance. See Example ??.

Sometimes probabilities that are not uniform can be seen as arising from a uniform probability space (in the sense of Definition ??, see Example ??)

Example 1.13: Sum of two dice

d:two dice sum

Consider the sample space of two dice

$$\Omega' = \{11, 12, \dots, 66\} = \{x_1 x_2 : x_1, x_2 \in \{1, \dots, 6\}\}$$

and endow it with the uniform probability. We will discuss in ?? the reason of this choice, but we anticipate that it is due to the fact that the dice are *fair* and they are rolled *independently*, see Subsection:ind?????. Assume we can only look at the sum of the results. This leads to a *coarse grained* sample space, see Example ?? where the elementary events, that is, the elements of the new sample space Ω , are the events described in Ω' by $\{(x_1, x_2) \in \Omega' : x_1 + x_2 = k\}$, and whose probabilities are

$$\mathbb{P}(A_2) = \mathbb{P}(A_{12}) = \frac{1}{36}$$

$$\mathbb{P}(A_3) = \mathbb{P}(A_{11}) = \frac{1}{18}$$

$$\mathbb{P}(A_4) = \mathbb{P}(A_{10}) = \frac{1}{12}$$

$$\mathbb{P}(A_5) = \mathbb{P}(A_9) = \frac{1}{9}$$

$$\mathbb{P}(A_6) = \mathbb{P}(A_8) = \frac{5}{36}$$

$$\mathbb{P}(A_7) = \frac{1}{6}.$$

sum

Therefore, the probability in Ω is not the uniform one.

Example 1.14: Extractions with replacement

ex:replace

Consider the sample space given by extractions with replacements, which are the object of Exercise 1.2.6. It makes sense to consider each outcome equally probable, that is

$$p_{(a,b)} = 1/9 \quad \text{for each } (a,b) \in \Omega.$$

e:extraction'replacement

This choice of a probability is a consequence of two properties, the fact that the extractions are *fair* and *independent*, see also ??? for a more general example. The fact that the extractions are independent is sometimes hidden in the extraction procedure: between two extractions we mix the balls.

Example 1.15: Extractions without replacement

ex:no'replacement

If in the extractions from the bowl of Example 1.14 and Exercise 1.15 we do not reinsert the ball the sample space is smaller or, equivalently, we assign $p_{rr} = p_{nn} = p_{gg} = 0$. Nonetheless the symmetry between the colours of the marbles suggests that we might regard each other outcome as equally probable. This is again a consequence of the two following facts, which will be discussed in a more general setting in (??). Each single extraction is fair and, since we mix the marbles before drawing again, the second extraction depends on the first one only because of the fact that the composition of the bowl changes.

Ex. 1.4.1 — A coin is tossed 4 times. Recall that the state space is $\Omega = \{0000, 0001, \dots, 1111\}$, where a 1 means that the associated coin gave heads. Suppose that the probability on Ω is uniform (we will see in Section ?? that this is due to independence of each coin toss). Calculate the probability of the events

1. The first toss gave 1
2. There are exactly two heads
3. There has been a head after a tail (that is, we observed a 01)
4. The event is obtained by fixing the first digit to 1 and allowing the other to take arbitrarily the values 0 or 1, that is, $E = \{1x_2x_3x_4, x_i \in \{0,1\}\}$. Since $|E| = 2^3$ and $|\Omega| = 2^4$, by formula (??) we obtain

$$\mathbb{P}(E) = 2^3/2^4 = \frac{1}{2}.$$

Ex. 1.4.2 — We throw two dice, and we suppose that the each output has the same probability (Section 1.5.2 will justify this choice with the hypothesis of independence and fairness of the dice)

1. Calculate the probability of obtaining a 2 and a 4

2. Calculate the probability of obtain 2 with the first die and 4 with the second

1.4.5 Basic properties

Let Ω be a sample space and $\mathbb{P}$ a probability defined on it. The first property tells us that there are no possible outcomes outside the ω_i .

p1

Proposition 1.13.

$$\mathbb{P}(\Omega) = 1 \quad (10)$$

e:normalization

Proof. It is a direct consequence of (10):

$$\mathbb{P}(\Omega) = \mathbb{P}(\{\omega_1, \omega_2, \dots, \omega_n\}) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_n) = 1.$$

□

We now formulate the *additivity* property of the probability.

p:monotone

Proposition 1.14 (Additivity). *Any two mutually exclusive events A, B , that is, events that satisfy $A \cap B = \emptyset$, verify*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B). \quad (11)$$

e:add

More generally,

p:exclusive

Proposition 1.15. *Any two events $A, B \subset \Omega$, it holds*

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

e:union

The proof looks more involved than what really is. The idea is simple and is the following up of the graphycal picture of the events as subset of the sample space using the Venn diagram in Figure 1.1. A probability can be seen as a measure of the areas covered by the events, and we can see that $\mathbb{P}(A) + \mathbb{P}(B)$ counts twice the area of $A \cap B$, while only once the area in $A \cup B$ that is not in $A \cap B$.

Proof. Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be the sample space. Without loss of generality, we assume that $A = \{\omega_1, \dots, \omega_s, \omega_{s+1}, \dots, \omega_l\}$, and that $B = \{\omega_1, \dots, \omega_s, \omega_{l+1}, \dots, \omega_{l+m}\}$. In this way, $A \cap B = \{\omega_1, \dots, \omega_s\}$, and $A \cup B = \{\omega_1, \dots, \omega_{l+m}\}$. Using the Definition 1.10,

$$\mathbb{P}(A) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s) + \mathbb{P}(\omega_{s+1}) + \dots + \mathbb{P}(\omega_l),$$

$$\mathbb{P}(B) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s) + \mathbb{P}(\omega_{l+1}) + \dots + \mathbb{P}(\omega_{l+m})$$

and

$$\mathbb{P}(A \cap B) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s).$$

In this way,

$$\mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s) + \mathbb{P}(\omega_{s+1}) + \dots + \mathbb{P}(\omega_{l+m}) = \mathbb{P}(A \cup B).$$

□

- **Corollary 1.16.** Let $A, B \subset \Omega$ be two mutually exclusive events, that is, two events such that $A \cap B = \emptyset$. Then

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B). \quad (12) \quad \boxed{\text{e:addc}}$$

Proof. Use (11) and note that $\mathbb{P}(A \cap B) = \mathbb{P}(\emptyset) = 0$. □

monotonicity

Corollary 1.17 (Monotonicity). Let A and B be two events $A, B \subset \Omega$. If $A \subset B$, then

$$\mathbb{P}(A) \leq \mathbb{P}(B).$$

The monotonicity is a very intuitive property. The probability that today's weather is bad is greater than the probability that today there is a storm, since if there is a storm there is also bad weather.

Proof. The intuition of the proof is that when we sum over the elementary events in B we are summing over the elementary events in A and the elementary events in $B \setminus A$. Therefore we write B as $A \cup (B \setminus A)$ and using (12) we obtain that $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$. □

complementary

Corollary 1.18. Let $A \subset \Omega$ be an event. The

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad \boxed{\text{e:compl}}$$

Proof. Write $\Omega = A \cup A^c$, apply (12) with $B = A^c$ and use (10). □

This property is also very natural. The probability that I will pass the test is 0.6. Thus, the probability that I will not pass the test is 0.4.

1.4.6 An Implementation Well Modelled by Probability

ss:r`prob1

In this section we give some possible

Given an n and some probabilities on elementary events $p_1, \dots, p_n$, see Definition ??, the computer is able to generate random outcomes with those probabilities. The following chunk of code, for instance, generates a natural number between 1 and 6 with probabilities proportional to p_{aux} , that is, with probabilities p .

```
n i- 6 # The number of elements of our sample space
p`aux i- c(1,2,3,0.5,1,5) # Here I take the $p_i$ to be positive numbers whose sum needs not to be 1
p i- p`aux/sum(p`aux) # Now I normalise them so that they sum 1
sample(1:n, 1, prob = p ) # The output of the sample

## [1] 5
```

This means that, defining $\omega_i = \text{"The output is } i \text{"}$, $\Omega = \{\omega_1, \dots, \omega_n\}$ and $p_1, \dots, p_n$ is a good model for the above output. Other systems for which the probability models work well that we have introduced before are Example ?? and the random extractions of balls from an urn in Example 1.16.

This is to be put later, after introducing independent sampling or maybe Extractions with or without replacement. The R command `sample` allows to implement the model of extractions with or without replacement.

1.5 CONDITIONAL PROBABILITY

In general, our evaluation of the probabilities changes if our state of information changes. If we observe that the sky is cloudy, our evaluation of the probability that it will rain will become greater. Given an event E , if $\mathbb{P}(\cdot)$ denotes the probability before the knowledge of E , we will denote the new probability, given the knowledge of E , by $\mathbb{P}(\cdot|E)$. With this notation, the previous example reads $\mathbb{P}(\text{"Today it will rain"}|\text{"It is cloudy"}) > \mathbb{P}(\text{"It will rain today"})$. In case the *conditioning* event E can be written in terms of the sample space, that is, in case $E \subset \Omega$, $\mathbb{P}(\cdot|E)$ can be described in simple terms.

1.5.1 Definition and first examples

Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be the sample space, $\mathbb{P}$ a probability over Ω , see Definition 1.11, and let $B \subset \Omega$ be an event. The probability *conditioned to* B is a new probability denoted by the symbol $\mathbb{P}(\cdot|B)$. Intuitively, it has to satisfy that

$$\mathbb{P}(\{\omega\}|E) = 0 \quad \text{if } \omega \notin E \quad \text{e:cond1}$$

and, since we do not have information on which event $\omega \in E$ caused E , we would like to maintain the proportions

$$\mathbb{P}(\{\omega\}|E) = C\mathbb{P}(\{\omega\}) = Cp_\omega, \quad \text{e:cond2}$$

for some constant C . The constant C is to be determined by (10):

$$1 = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}|E) = \sum_{\omega \in E} C\mathbb{P}(\{\omega\}) = C\mathbb{P}(E) \quad (13) \quad \text{e:cond3}$$

so that $C = \mathbb{P}(E)$. For the sake of concreteness we rewrite (13) using a different and extended notation. Let $E = \{\omega_{i_1}, \dots, \omega_{i_k}\}$, for some $k \in \mathbb{N}$ and $1 \leq i_1 < \dots < i_k \leq n$ be an event, then

$$\begin{aligned} 1 &= \mathbb{P}(\{\omega_1, \dots, \omega_n\}|\{\omega_{i_1}, \dots, \omega_{i_k}\}) \\ &= \sum_{i=1}^n \mathbb{P}(\{\omega_i\}|\{\omega_{i_1}, \dots, \omega_{i_k}\}) = C \sum_{j=1}^k \mathbb{P}(\{\omega_{i_j}\}) = C\mathbb{P}(E) \end{aligned} \quad \text{e:cond4}$$

Therefore $C = \frac{1}{\mathbb{P}(E)}$ and the probability of an ev

that is, we denote the probability of A conditioned to B by $\mathbb{P}(A|B)$. Of course, the new probability has to satisfy $\mathbb{P}(\omega|B) = 0$ if $\omega \notin B$. If $\omega \in B$, what should the new value $\mathbb{P}(\omega|B)$ be? Since the new information tells us only that B is true, but it does not give us information about the single ω s in B , the proportions given by $\mathbb{P}$ in B should be maintained by $\mathbb{P}(\cdot|B)$. Therefore we set $\mathbb{P}(\omega|B) = c\mathbb{P}(\omega)$ if $\omega \in B$, for some constant c . c which

can be determined by requiring $\mathbb{P}(\cdot|B)$ has to satisfy (??). If $B = \{\omega_1, \dots, \omega_k\}$, we obtain that

$$\begin{aligned} 1 &= \mathbb{P}(\omega|B) + \dots + \mathbb{P}(\omega_k|B) + \mathbb{P}(\omega_{k+1}|B) + \dots + \mathbb{P}(\omega_n|B) \\ &= c\mathbb{P}(\omega_1) + \dots + c\mathbb{P}(\omega_k) = c(\mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_k)) = c\mathbb{P}(B). \end{aligned}$$

Therefore $c = 1/\mathbb{P}(B)$ and

$$\mathbb{P}(\omega|B) = \begin{cases} \frac{\mathbb{P}(\omega)}{\mathbb{P}(B)} & \text{if } \omega \in B \\ 0 & \text{if } \omega \notin B, \end{cases}$$

e:condi

Considering an event $A = \{\omega_1, \dots, \omega_s, \omega_{k+1}, \dots, \omega_{k+m}\}$, where $s \leq k$ formula (1.10) leads to

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(\omega_1) + \dots + \mathbb{P}(\omega_s)}{\mathbb{P}(B)} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

conditional

Definition 1.19. Let Ω be the sample space and let $\mathbb{P}$ be a probability. Consider an event $B \subset \Omega$ such that $\mathbb{P}(B) > 0$. The conditional probability $\mathbb{P}(\cdot|B)$ given B is the probability measure defined by

$$\Omega \ni A \mapsto \mathbb{P}(A|B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)},$$

that is, is the probability that associates to each $A \subset \Omega$ the value $P(A|B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$.

As shown in Figure ??, there is an easy interpretation of the conditional probability. Recall from ??? that one can interpret a probability as a way of measuring areas (in the discrete setting as a way to count which gives different weights to different events). If we observe B , our new evaluation of the area doesn't change in E and outside E is updated set zero. Since one of the conditions imposed to a probability is that the total area is 1, see Proposition ??, to maintain this property we need to divide the new evaluation by E . In this sense we maintain the proportions of our evaluations inside E (since we do not have any information on which event caused E) and we set them to 0 outside, since we know that E happened. Alternative, one can think that we are redistributing the mass of E^c , that is $\mathbb{P}(E^c)$, to E , in such a way that the proportions do not change. One can think at a probability as a distribution of a mass, which in total is 1, over Ω . If one takes the distribution of mass given by $\mathbb{P}$ and sets the mass of B^c to be zero, one has a distribution of mass whose total is not 1. In order to obtain the distribution of mass given by the conditional probability $\mathbb{P}(\cdot|B)$, one has therefore to redistribute the mass it has removed, $\mathbb{P}(B^c)$, in such a way to not change the relative masses of the singletons in B . Thus $\mathbb{P}(\omega|B) = \mathbb{P}(\omega) + \mathbb{P}(\omega)\mathbb{P}(B^c)/\mathbb{P}(B)$. This way of thinking can be useful for the Exercise ??. Note also that Example (1.5.1) and Exercise (??) show that probabilities can vary a lot if conditioned to the an event B . This is schematically shown in Figure (1.3)

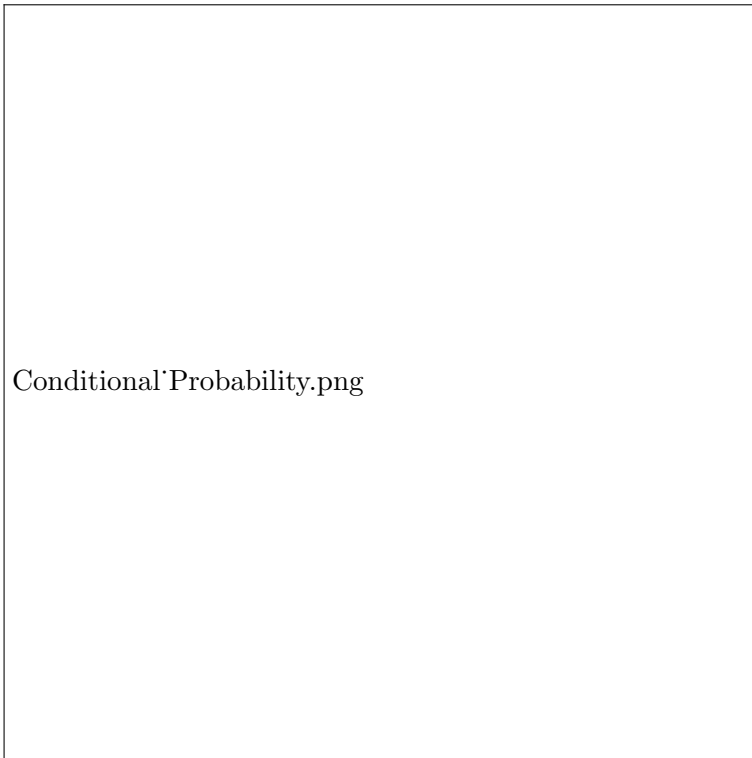
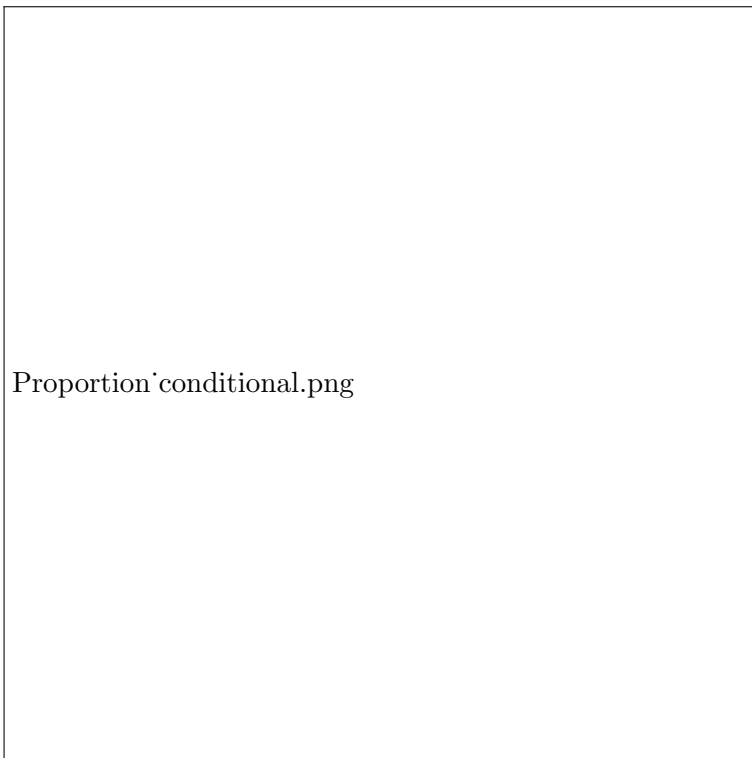


Figure 1.2: We forget about B^c and we consider $\mathbb{P}$ restricted to subsets of $\mathbb{B}$. However, to obtain total mass 1 without favouring any of the $\omega \in B$, we multiply by $\frac{1}{\mathbb{P}(B)}$.

f:cond`prob



Proportion`conditional.png

Figure 1.3: $\mathbb{P}(E)$ is much greater than $\mathbb{P}(E|B)$.

fig:prop

^j In other words, on Ω we assume that $\mathbb{P}$ is the uniform one. Why this is the case investigated in Subsection 1.5.2

Ex. 1.5.1 — We throw two dice and assume that each of numbers is equally probable^j what is the probability conditioned on "The sum of the two dice is 7"?

Ex. 1.5.2 — You toss a fair coin independently 4 times¹. Show that, knowing that only one head has been observed, the head is equally probable to have appeared in the 1st, 2nd, 3rd and 4th position.

Answer (Ex. 1.5.2) — The sample space is $\Omega = \{0000, 0001, 0010, \dots, 1111\}$. Let $B = \{x_1x_2x_3x_4, x_1 + x_2 + x_3 + x_4 = 1\} = \{1000, 0100, 0010, 0001\}$ be the event that only one heads result has shown up. The probability of the singleton $x_1x_2x_3x_4$, where $x_i \in \{0, 1\}$ for $i = 1, 2, 3, 4$, conditioned to B is given by

$$\mathbb{P}(x_1x_2x_3x_4|B) = 0$$

if $x_1 + x_2 + x_3 + x_4 = 1$ and is equal to

$$\mathbb{P}(x_1x_2x_3x_4|B) = \frac{1/2^4}{4/2^4} = 1/4,$$

which is to say that there is not favourite position in which the heads might appear.

Ex. 1.5.3 — Let $\Omega = \{\omega_1, \dots, \omega_n\}$ be a sample space and let $\mathbb{P}$ be a probability assigned to it. Let $B \subset \Omega$ be an event, and assume $\mathbb{P}(B) > 0$.

1. Prove that $\mathbb{P}(\cdot|B)$ is a probability on the sample space B .
2. Assume that $\mathbb{P}$ is the uniform probability defined in (??) on Ω . Show that $\mathbb{P}(\cdot|B)$ is the uniform probability on the new sample space B .

Answer (Ex. 1.5.3) — 1. In order to show that $\mathbb{P}(\cdot|B)$ is a probability measure, we have to show that

- $\mathbb{P}(\omega|B) \in [0, 1]$
- ((??) holds) $\sum_{\omega \in B} \mathbb{P}(\omega|B) = 1$
- ((1.10) holds) For each $F \subset B$, $\mathbb{P}(F|B) = \sum_{\omega \in F} \mathbb{P}(\omega|B)$.

The first condition is true since, if $\omega \in B$, $\mathbb{P}(\omega) \leq \mathbb{P}(B)$, and therefore $\mathbb{P}(\omega|B) = \mathbb{P}(\omega)/\mathbb{P}(B) \leq 1$. The second condition, which can be rewritten by Proposition (1.13) as

$$\mathbb{P}(B|B) = 1$$

is true since

$$\mathbb{P}(B|B) = \frac{\mathbb{P}(B \cap B)}{\mathbb{P}(B)} = 1.$$

¹ The same remark applies here: the probability is the uniform one and in Subsection 1.5.2 we will see why

The last condition is true since

$$\begin{aligned}\mathbb{P}(F|B) &= \frac{\mathbb{P}(F \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(F)}{\mathbb{P}(B)} \\ \sum_{\omega \in F} \frac{\mathbb{P}(\omega)}{\mathbb{P}(B)} &= \sum_{\omega \in F} \mathbb{P}(\omega|B).\end{aligned}$$

2. As for the second question, since we have just proved that $\mathbb{P}$ is a probability measure on B , it suffices to note that every elementary event has the same probability (recall how we have derived formula (??): simply by assuming that the probability was equal on each elementary event). In fact $\forall \omega \in B \mathbb{P}(\omega|B) = \mathbb{P}(\omega)/\mathbb{P}(B)$ does not depend on ω , since by assumption $\mathbb{P}$ does not depend on ω .

Ex. 1.5.4 — ([Feller], Chapter V.8) Three dice are rolled. If no two show the same face, what is the probability that one is an ace?

Answer (Ex. 1.5.4) — The sample space is $\Omega = \{(i, j, k), i, j, k \in \{1, \dots, 6\}\}$, and the event "No two dice show the same face" is $A = \{(i, j, k) \in \Omega, i \neq j, j \neq k, i \neq k\}$. The event B = "The sum of the two dice is 7" is such that $A \cap B$ is the disjoint union of the events "A and the 1 is in 1st position", "A and the 1 is in 2nd", "A and the 1 is in 3rd position". This union is disjoint since if A happens, a 1 in the 1st position implies that there is no 1 neither in the 2nd nor in the 3rd. In formulas $A \cap B = (A \cap B_1) \cup (A \cap B_2) \cup (A \cap B_3)$, where $B_1 = \{(1, j, k) \in \Omega\}$, $B_2 = \{(i, 1, k) \in \Omega\}$ and $B_3 = \{(i, j, 1) \in \Omega\}$. Thus

$$\begin{aligned}\mathbb{P}(B|A) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \cap B_1)}{\mathbb{P}(A)} \\ &+ \frac{\mathbb{P}(A \cap B_2)}{\mathbb{P}(A)} + \frac{\mathbb{P}(A \cap B_3)}{\mathbb{P}(A)} = 3 \frac{\mathbb{P}(A \cap B_1)}{\mathbb{P}(A)}.\end{aligned}$$

The event $A \cap B_1 = \{(1, j, k), j \neq k\}$ is composed by $6 \times 5 = 30$ elements. The event A is composed by $6 \times 5 \times 4$ elements. Thus

$$\mathbb{P}(B|A) = \frac{6 \times 5 / 216}{6 \times 5 \times 4 / 216} = \frac{6 \times 5}{6 \times 5 \times 4} = 1/4$$

Ex. 1.5.5 — Consider an urn containing 2 blue marbles, 1 green marble and 1 red marble. Draw randomly (with uniform probability) one of them.

1. What is the probability of extracting a red marble if you know that the extracted marble is not green?
2. We do the following procedure. We draw a marble uniformly. If it is red we draw another marble independently, if not we finish the procedure. We are interested in the colour of the extracted marble: what is the probability of extracting a red marble?
3. Note that it is not necessary to know the probability of drawing in one draw the red ball, that is $\mathbb{P}(r)$. Note also that in the second exercises

gives you a procedure to draw from a bowl where you have removed the red marbles, even if you are drawing from a bowl that has them.

Answer (Ex. 1.5.5) — 1. The sample space is $\Omega = \{r, b, g\}$, $\mathbb{P}(r) = 1/4$, $\mathbb{P}(b) = 1/2$ and $\mathbb{P}(g) = 1/4$. Thus

$$\mathbb{P}(g^c | r^c) = \frac{\mathbb{P}(b)}{\mathbb{P}(b, g)} = \frac{2}{3}$$

2. Now the sample space is more complicated, and resembles the game of Exercise (1.2.12)

$$\Omega = \{r \dots r c_n, n \in \mathbb{N}, c_n \in \{b, g\}\} \cup \{rr \dots\}$$

The probability that we place in Ω is the following: The infinite sequence has probability 0: $\mathbb{P}(r \dots)$, while the finite sequences have probability

$$\begin{aligned} \mathbb{P}(rr \dots r c_n) &= \mathbb{P}(rr \dots r c_n | \text{"first extraction } r") \mathbb{P}(r) = \mathbb{P}(r \dots c_{n-1}) \mathbb{P}(r) \\ &= \dots = \mathbb{P}(r)^{n-1} \mathbb{P}(c_n) \end{aligned}$$

(It will be maybe more clear once the geometric random variable has been obtained) and we take $\mathbb{P}(c) = (\text{number of balls of colour } c) / (\text{total number of balls})$. The event I draw a blue ball is the event $\{b, rb, rrb, \dots\}$ which has probability

$$\begin{aligned} \mathbb{P}(\text{"I draw a blue ball"}) &= \mathbb{P}(b) + \mathbb{P}(b)\mathbb{P}(r) + \mathbb{P}(b)\mathbb{P}(r)^2 + \dots \\ &= \mathbb{P}(b) \left(\sum_{i=0}^{+\infty} \mathbb{P}(r)^i \right) = \frac{\mathbb{P}(b)}{1 - \mathbb{P}(r)} = \frac{\mathbb{P}(b)}{\mathbb{P}(b) + \mathbb{P}(g)} = 2/3 \end{aligned}$$

3. The above procedure can be used to draw on a bowl of. In a more realistic situation, maybe you want to interview families with children, but you don't have at disposal the "bowl", that is, the list with of the families of two children. This exercises tells that it suffices to contact each family until a family with children is contacted. In more complex situations (i.e. in montecarlo simulations), a similar method is used to get to know the conditional probabilities

Example 1.16: Sampling interpretation of the Conditional Probability

Suppose that the Italian population is composed by N individuals, that N_B of them are taller than 175 cm and that N_A of them come from Sardinia. Let the events that a person chosen at random (i.e. with uniform probability) is taller than 175 cm and a comes from Sardinia be A and B , respectively. Then $\mathbb{P}(A) = N_A/N$ and $\mathbb{P}(B|A) = N_{AB}/N_A$, where N_{AB} is the number of people coming from Sardinia being taller

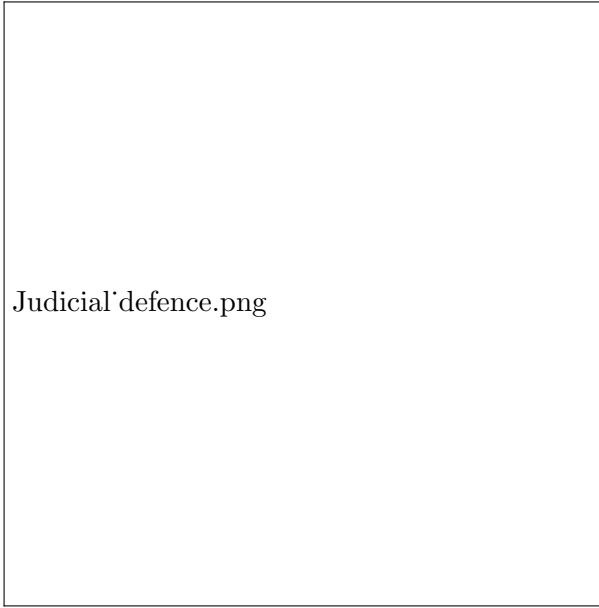


Figure 1.4: Defense argument: The proportion between the two areas is small

than 175 cm. The interpretation of the probability measure $\mathbb{P}(\cdot|A)$ is clear: we are extracting a random person from the subpopulation N_A (which is a population on its own right), the one composed by people from Sardinia .

e:Judicial

A judicial case A woman had been killed, and the principal suspect was his husband. During the investigations, the police discovered that the husband had beaten his wife more than once. His lawyer used some data that showed that only 1/10000 men beating their wives end up killing them, thus claiming that there were no sufficient clues to judge him guilty. Fortunately, the prosecutors pointed a fallacy in the defense argument. 1/10000 gives an estimate of the probability that a man kills his wife knowing that he beats her. Here much more is known: the wife actually died.

Denoting the events

A = "The husband beat the wife "

B = "The woman has been assassinated by his husband "

C = "The woman has been assassinated by a person different from her husband "

The claim of the defence is depicted in Figure 1.4 We wish to compute $\mathbb{P}(B|A \cap (B \cup C))$, which, which is depicted in Figure 1.5.

$$\begin{aligned} \mathbb{P}(B|A \cap (B \cup C)) &= \frac{\mathbb{P}(B \cap A \cap (B \cup C))}{\mathbb{P}(A \cap (B \cup C))} = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}((B \cup C) \cap A)} \\ &= \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(B \cap A) + \mathbb{P}(C \cap A)} = \frac{1}{1 + \frac{\mathbb{P}(C \cap A)}{\mathbb{P}(B \cap A)}}. \end{aligned} \quad (14) \quad \text{e:proportions}$$

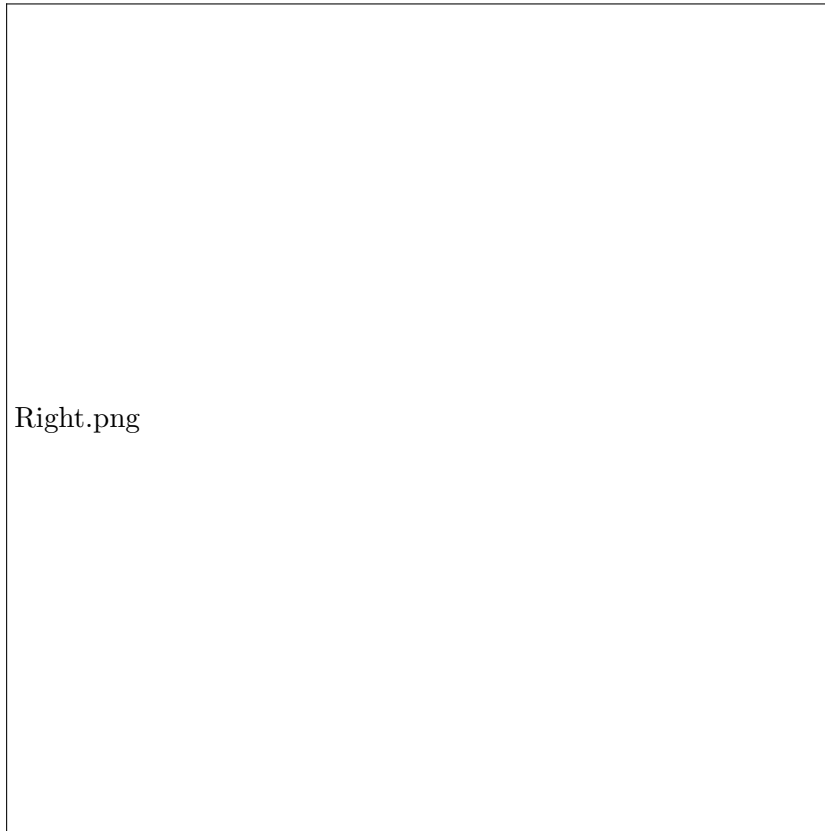


Figure 1.5: What we need to compute: the proportion of the blue area with respect to the total area. Why this is what we need to compute is shown in (14)

fig:right

Thus it is not $\mathbb{P}(B|A)$ that matters, but only its relative value $\mathbb{P}(B|A)/\mathbb{P}(C|A) = \mathbb{P}(C \cap A)/\mathbb{P}(B \cap A)$, as shown in Figure 1.5. We assume that C and A are independent, so that $\mathbb{P}(C|A) = \mathbb{P}(C)$ and (14) becomes

$$\mathbb{P}(B|A \cup (B \cap C)) = \frac{1}{1 + \frac{\mathbb{P}(C)}{\mathbb{P}(B|A)}}.$$

We estimate the $\mathbb{P}(C)$ by the probability that a woman gets killed: $\mathbb{P}(C) \leq \mathbb{P}(B \cup C)$ which they estimated by $\mathbb{P}(B \cup C) = 1/100000$. Therefore

$$\frac{\mathbb{P}(C|A)}{\mathbb{P}(B|A)} \leq 1/10.$$

Plugging in this value, one obtains that

$$\mathbb{P}(B|A \cap (B \cup C)) \geq 10/11$$

At the end, the man has been judged guilty.

1.5.2 Independence of Events

ss:independence

In view of the definition of conditional probability in Definition ?? the following definition is natural to say that the events A and B are independent if $\mathbb{P}(A) = \mathbb{P}(A|B)$. In a more symmetric form

Definition 1.20. Two events $A, B \subset \Omega$ are *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

e:ind

It is important to notice that the word “independent” can be misleading, since it could be wrongly associated to the absence of a causal relation between A and B . Example ?? shows that there are events not related by any causal relation that are dependent and that, conversely, there are independent events that are causal. The events A and B are independent if from the knowledge of one of them you cannot infer anything on the evaluation of the probability of the other one: the probability remains the same. For this reason the independence property is sometimes stochastic independence. Another way to see that the notions of causality and independence are unrelated is by noting that the notion of independence depends from the probability $\mathbb{P}$, while the notion of causality does not.

Note that the notion of independence is unrelated with the notion of causality. The notion of causality is a notion that takes into account only the relation between events, while the notion of independence takes into account the notion of probability. Independence means that, given $\mathbb{P}$, that is, having an evaluation of the probabilities of the events, we can infer no information from the realization of B that causes to change the evaluation of the probability of A . Thus is a probability dependent hypothesis and, for instance, Exercise (??) shows that coin tosses are not independent if we don’t know exactly the parameter of the coin, since we can infer some information useful to determine the coin parameter by tossing the coin, which, in turn, will change the probability of the next toss. It is the same situation of (1.5.13), which makes clear the striking fact that the number of shark attacks and the number of ice creams sold are numbers with a strong dependence, even if, of course, not related by any causal relation.

Example 1.17: Two independent uniform samplings

Consider a bowl where 3 marbles, 1 red, 1 blue and 1 green, are placed and draw two marbles with replacement. The state space of the system is

$$\Omega = \{(r, r), (r, b), \dots, (g, g)\},$$

We want to prove that the following are equivalent:

- The probability measure on Ω is the uniform one.
- In each single extraction you can draw each marble with the same probability and the extractions are independent

To see what the second condition means precisely, let’s assume that $\mathbb{P}$ is uniform and define the events

$$F_c = \text{“The colour of the first marble is } c\text{”} = \{(c, r), (c, b), (c, g)\},$$

for $c = r, b, g$, and

$$S_c = \text{"The colour of the second marble is } c" = \{(r, c), (b, c), (g, c)\},$$

for $c = r, g, b$. By saying that each extraction is a "random" extraction (that is, is uniform) we mean that

$$\begin{aligned}\mathbb{P}(F_c) &= 1/3 \\ \mathbb{P}(S_c) &= 1/3\end{aligned}\tag{15}$$

1

for each $c = b, g, r$, which is easily seen to hold true. By the fact that the extractions are independent we mean that for each c and c' in $\{r, b, g\}$

$$\mathbb{P}(F_c \cap S_{c'}) = \mathbb{P}(F_c)\mathbb{P}(S_{c'}).\tag{16}$$

2

The event on the left hand side is the singleton $\{(c, c')\}$, so that its probability is $1/9$. The events on the right hand side have probability $1/3$, so that the above equality holds. Let's assume that now (15) and (16) hold. Then

$$\begin{aligned}\mathbb{P}(c, c') &= \mathbb{P}(F_c \cap S_{c'}) \\ &= \mathbb{P}(F_c)\mathbb{P}(S_{c'}) = 1/3 \times 1/3 = 1/9,\end{aligned}$$

so that $\mathbb{P}$ is uniform. This example will be generalised in Section ?? to the case where the two extraction procedure can be performed jointly in an arbitrary way. Examples (1.18) and Exercise (??) and (??) will show examples in which you perform two dependent extractions and how this can lead to surprising results.

Example 1.18: Mouty Hall Problem

MoutyHall

In a game show called the Mouty Hall, the participants had to choose between three doors. Behind 1 door there was a car while behind the other two there was a goat. The participant had to guess which door contained the car. After the first choice, the host disclosed a door which had behind a goat and then asked the participant whether or not he would like to change the choice of the door. The result is at a first sight surprising. Let's call Strategy 1 the strategy in which we don't change the door, and Strategy 2 the strategy in which we decide to change the door. Then $\mathbb{P}(\text{"win"}) = 1/3$ if we play following Strategy 1, and $\mathbb{P}(\text{"win"}) = 2/3$ if we play by Strategy 2. To prove the above fact, assume that the participant choses the door 1 and that he plays following Strategy 2. The result is clear if one observes that the participant wins if and only if he chooses a goat at

the beginning. **Exercise:** Consider the Mouny Hall problem with n doors and k cars. What is the probability of winning under Strategy 2?

Solution The sample space should encode the doors, the thing that is behind the door, the first door chosen by the participant and its second choice. However, we can give a reduced description since we are interested only on the choices the participant makes. The sample space can be taken to be $\Omega = \{CC, GG, GC, CG\}$ where the first digit denotes what is there behind the first door and similarly for the second. Let's denote the event "The first choice is a car" by F , and the event "The second choice is a car" = "win" by S . Therefore

$$\mathbb{P}(S) = \mathbb{P}(S|F)\mathbb{P}(F) + \mathbb{P}(S|F^c)\mathbb{P}(F^c),$$

but, given F , after the goat is revealed, the participant chooses between $n-2$ doors with $k-1$ cars. Given F^c , then, after the host reveals the goat, has to choose between $n-2$ doors and k cars. Therefore

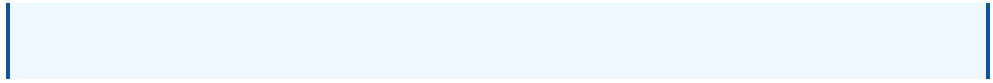
$$\begin{aligned}\mathbb{P}(S) &= \mathbb{P}(S|F)\mathbb{P}(F) + \mathbb{P}(S|F^c)\mathbb{P}(F^c) \\ &= \frac{k-1}{n-2} \frac{k}{n} + \frac{k}{n-2} \frac{n-k}{n}\end{aligned}$$

So, it is not difficult to understand the Mouny Hall problem: what happens is that the different extractions, the first door and the second one are dependent.

The next "paradox" has interest on its own and shows that it is important to clearly state the conditioning event

Example 1.19: Boy or Girl Paradox

Consider a family with two children and consider their genders. The state space is $\Omega = \{MM, MF, FM, FF\}$ and consider the uniform probability on it.^k



1. What is the probability that the second child of a couple is a boy, knowing that the first one is a boy?
2. Assume that we get to know that one of the children is a boy. What is the probability that both children are males?
3. We have chosen a child randomly (and independently on the gender, of course) and have seen that he is male. What is the probability that both children are boys? (Note that the first point is a particular case of this one, when the chosen child is surely the first one).

For the first question, look at the sample space $\Omega = \{(M, M), (M, F), (F, M), (F, F)\}$ and endow it with the uniform probability (see Example 1.17). The event $A = \text{"The first child is male"}$ is given by $\{(M, M), (M, F)\}$. A basic computation shows that

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{2/4} = 1/2.$$

As for the second question, maybe counterintuitively, the answer is $1/3$. In this case the conditioning event A is $\{(M, M), (M, F), (F, M)\}$, and

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{3/4} = 1/3.$$

In the last case, the probability is $1/2$! Let's take as state space $\Omega = \{MM1, MM2, MF1, MF2, FM1, FM2, FF1, FF2\}$, where the last number means that you have observed the gender of the child 1 or child 2. The event $A = \text{"you observed a male child"} = \{MM1, MM2, MF1, FM2\}$, while the event "The couple has two male children" is $B = \{MM1, MM2\}$. Since we assume that the choice is done independently, we have that if we denote $1 = \text{"I observed the first child"}$ $\mathbb{P}(MM1|1) = \mathbb{P}(MF1|1) = \mathbb{P}(FF1|1) = 1/4$. Similarly for the event $2 = \text{"I observed the second child"}$. We assume that $\mathbb{P}(1) = p$ and we compute

$$\mathbb{P}(A) = \mathbb{P}(\{MM1, MF1\}|1)\mathbb{P}(1) + \mathbb{P}(\{MM2, FM2\}|2)\mathbb{P}(2) = p/2 + (1-p)/2 = 1/2,$$

and

$$\mathbb{P}(\{MM1, MM2\}) = \mathbb{P}(MM1|1)\mathbb{P}(1) + \mathbb{P}(MM2|2)\mathbb{P}(2) = 1/4.$$

$$\mathbb{P}(\{MM1, MM2\}|A) = \mathbb{P}(A \cap B)/\mathbb{P}(A) = 1/2.$$

Even if the second point of the example might come as a surprise, if one interprets the probability with a wrong frequentist definition, if one were to obtain the list of all the families with two children such that one is a male, then approximately $1/3$ of the families in those lists

will have 2 male children. The procedure 3 would have the following equivalent in the frequentist approximation. From the list of all the families with two children, you toss an unfair coin to decide which children to take into account and to check. So there is a

Ex. 1.5.6 — You toss a fair coin 4 time. Assume that every outcome is equally probable (we will rigorously see why). Show that the events "The fourth toss gave heads" and the event "The first three tosses gave head" are independent.

Ex. 1.5.7 — In the setting of Example 1.17 you draw two marbles without replacement.

1. What are the probabilities of each single extraction?
2. Are extractions independent?

ex:3indep

Ex. 1.5.8 — $\Omega = \{1, 2, 3, 4\}$ and $\mathbb{P}(i) = 1/4$ for every $i \in \Omega$. Show that $A = \{1, 2\}$ $B = \{1, 3\}$ and $C = \{2, 3\}$ are pairwise independent but not independent.

Ex. 1.5.9 — 3 prisoners, (A,B,C) are condemned to death. They decide to save one of them, but the name will be given only some minutes before the executions. A asks to a guard if he will be saved, but gets no answer. A asks then which one between B and C will be executed, and the guard says C. At this point A says. "Now only one between me and B will be executed; I have 50% of chances to survive ". Is it correct his reasoning?

Ex. 1.5.10 — There are three cards one of them has both faces black, one with both red faces, and the last one with one black face and the other face red. You draw a card and observe a red face. What is the probability that the other face is red?
This example shows that is important to clearly state the conditioning event.

1. What is the probability that the second child of a couple is a boy, knowing that the first one is a boy?
2. We know that one of the child is a boy. What is the probability that both children are males?
3. We have chosen a child randomly (and independently on the gender, of course) and have seen that he is male. What is the probability that both children are boys? (Note that the first point is a particular case of this one, when the chosen child is surely the first one).

For the first question, look at the sample space $\Omega = \{(M, M), (M, F), (F, M), (F, F)\}$ and endow it with the uniform probability (in section 1.16 we will see that this is due to the fact that we regard the gender of the two child as independent, and the probability of each single child to be either a boy or a girl $1/2$).

The event $A = \text{"The first child is male"}$ is given by $\{(M, M), (M, F)\}$. We are interested in

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{2/4} = 1/2.$$

Maybe counterintuitively, the answer is $1/3$. But this is clear since the conditioning event A is $\{(M, M), (M, F), (F, M)\}$, and

$$\mathbb{P}((M, M)|A) = \mathbb{P}(M, M)/\mathbb{P}(A) = \frac{1/4}{3/4} = 1/3.$$

Using a frequentist approach, if one had the list of all the couples with two children, one of which male, approximately $1/3$ would have two male children.

In the last case, the probability is $1/2$! Let's take as state space $\Omega = \{MM1, MM2, MF1, MF2, FM1, FM2, FF1, FF2\}$, where the last number means that you have observed the gender of the child 1 or child 2. The event $A = \text{"you observed a male child"} = \{MM1, MM2, MF1, FM2\}$, while the event $B = \text{"The couple has two male children"} = \{MM1, MM2\}$. Since we assume that the choice is done independently, we have that if we denote $1 = \text{"I observed the first child"}$ $\mathbb{P}(MM1|1) = \mathbb{P}(MF1|1) = \mathbb{P}(FF1|1) = 1/4$. Similarly for the event $2 = \text{"I observed the second child"}$. We assume that $\mathbb{P}(1) = p$ and we compute

$$\mathbb{P}(A) = \mathbb{P}(\{MM1, MF1\}|1)\mathbb{P}(1) + \mathbb{P}(\{MM2, FM2\}|2)\mathbb{P}(2) = p/2 + (1-p)/2 = 1/2,$$

and

$$\mathbb{P}(\{MM1, MM2\}) = \mathbb{P}(MM1|1)\mathbb{P}(1) + \mathbb{P}(MM2|2)\mathbb{P}(2) = 1/4.$$

$$\mathbb{P}(\{MM1, MM2\}|A) = \mathbb{P}(A \cap B)/\mathbb{P}(A) = 1/2.$$

This apparent paradox is due to the fact that the event A is different from the event $B = \text{"There is at least one male child"}$. If you repeat the same calculations with B instead of A , you should obtain $1/3$.

You can interpret this example as the fact that it is important to know how to get the information and it is important to specify the conditioning event rigorously.

Definition 1.21. Three events A, B and C are independent if they are pairwise independent

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B),$$

$$\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C),$$

$$\mathbb{P}(B \cap C) = \mathbb{P}(B)\mathbb{P}(C),$$

and

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

An example three events that are pairwise independent but not independent is given in Exercise 1.5.8.

1.5.3 Sampling Interpretation of Conditional Probability

ss:r'cond

We now come back to the example of Subsection ??, where $\Omega = \{\omega_1, \dots, \omega_6\}$ and we assume that we observe $E = \{\omega_1, \omega_4, \omega_5, \omega_6\}$. We can define the conditional probability by setting $p_2 = p_3 = 0$ and normalising the thus obtained vector so that the sum is 1.

Looking at Example ??, if we know how to sample from $\mathbb{P}$, and indeed we were able to do it in Subsection ??, we can sample from the $\mathbb{P}()$

1.5.4 Total probability Formula

When speaking about events many times one reasons by cases in the following sense. Assume you don't know whether to go out this night, then, in order to think at the event "I will go out this night" one thinks at this event first in the case it rains, then in the case it is good weather. Since the probability is simply a quantitative version of the logic we use when speaking about events, we might hope to reproduce the same reasoning at the quantitative level of probabilities. This is indeed true and it is done in the Total Probability Formula.

Proposition 1.22 (Total Probability Formula for two conditioning events). *Let $B \subset \Omega$ be an event such that $\mathbb{P}(B) \in (0, 1)$. Then*

$$\mathbb{P}(A) = \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c) \quad (17)$$

e:totaleasy

Proof.

$$\mathbb{P}(A) = \mathbb{P}(A \cap \Omega) = \mathbb{P}(A \cap (B \cup B^c)) = \mathbb{P}((A \cap B) \cup (A \cap B^c)).$$

Since $A \cap B$ and $A \cap B^c$ are disjoint, we can use the (11) to develop the above expression into

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} + \frac{\mathbb{P}(A \cap B^c)}{\mathbb{P}(B^c)} \\ &= \mathbb{P}(A|B)\mathbb{P}(B) + \mathbb{P}(A|B^c)\mathbb{P}(B^c), \end{aligned}$$

which is what we wanted to prove. $\square$

Ex. 1.5.11 — We toss a fair coin. If the coin gives heads we draw a ball from a bowl that contains a blue ball, a red ball and a white ball. If the coin gives tails, we draw a ball from a bowl that contains a blue ball or a red ball. What is the probability of drawing a red ball?

Answer (Ex. 1.5.11) — Let's call B ="The coin gave heads". We know that $\mathbb{P}(B) = \mathbb{P}(B^c) = 1/2$. We also know that the event r =" We draw a red ball" has the following probabilities conditioned on B :

$$\mathbb{P}(r|B) = 1/3, \quad \mathbb{P}(r|B^c) = 1/2.$$

We can apply formula (17) to obtain

$$\mathbb{P}(r) = \mathbb{P}(r|B)\mathbb{P}(B) + \mathbb{P}(r|B^c)\mathbb{P}(B^c) = 1/2 \cdot 1/3 + 1/2 \cdot 1/2 = 5/12.$$

We now give the total probability formula in full generality. The proof is almost the same, but we need to introduce the concept of partition of the sample space Ω :

Definition 1.23. The events $B_1, \dots, B_n$ are said to be mutually disjoint if $B_i \cap B_j = \emptyset$ whenever $i \neq j$. A partition is a set $\{B_1, \dots, B_n\}$ of mutually disjoint events B_i , such that $\Omega = \bigcup_{i=1}^n B_i$.

Examples of partitions are

- In 4 coin tosses, a partition of $\Omega = \{0000, 0001, \dots, 1111\}$ in two events is, for example, $\{B_1, B_2\}$, where B_1 ="The first toss gave tail" and $B_2 = (B_1)^c$ ="The first toss was head". Another partition would be $\{C_0, C_1, C_2, C_3, C_4\}$ where C_i = "There have been i heads".
- In the roll of a die, an example of partition of $\Omega = \{1, 2, 3, 4, 5, 6\}$ is $\{B_1, B_2\}$, where $B_1 = \{2, 4, 6\}$ = "The result is an even number", and $B_2 = \{1, 3, 5\}$ = "The result is an odd number".
- If we roll two dice, a partition is $\{B_2, \dots, B_{12}\}$, where B_i is the event "The sum of the dice is i ".
- We observe some coin tosses but we don't know how many of them there will be. Thus Ω is the one given by (??) and it can be partitioned into $\Omega = \bigcup_n \Omega_n$, where Ω_n =" There have been n tosses " = $\{x_1 \dots x_n, x_i \in \{0, 1\}\}$.

In other words a partition is a family of mutually disjoint events (or hypotheses) that together they include all the possible eventualities, conditions intuitive and necessary to make reasonings of the form " If B_1 , then and you can reason separately on each element of the partition, both at the level of sets and at the level of probabilities thanks to the next Note that if a partition consists of two events B_1 and B_2 , then $B_2 = (B_1)^c$.

Proposition 1.24 (Total probability formula). *Let $\{B_1, \dots, B_n\}$ be a partition of Ω . Then*

$$\mathbb{P}(A) = \mathbb{P}(A|B_1)\mathbb{P}(B_1) + \dots + \mathbb{P}(A|B_n)\mathbb{P}(B_n).$$

We omit the prove, which is similar to the previous and we remark that, up to some basic knowledge of series, one could take a partition to be an infinite family of the form $\{B_i\}_{i \in \mathbb{N}}$, and the total probability formula would still be true.

Ex. 1.5.12 — Consider an infection that is contagious and which is suffered by 1 person out of 1000. There is a test used to determine whether a person has the infection and the test has a 5% of possibility of giving a false positive: You don't have the infection but the test is positive, while the probability of having a false negative is zero.

1. If you are tested positive, what is the probability of actually having the infection?
2. You have some symptoms and the doctors estimate around 50% of probability that you have the infection. What is the probability that you have the infection provided that the test gives a positive result?

ex:indep

Ex. 1.5.13 — We have two urns, one containing 3 red balls, 1 blue ball, and one white ball, the other one containing 3 blue balls, 1 red ball and 1 white ball. We toss a fair coin to choose one of the two urns.

1. We draw a ball from the chosen urn. What is the probability that you extract a red ball? (Hint: Use the total probability formula)
2. Now we extract a ball, reinsert it in the urn, and draw again another ball from the same urn. Are the event "The first ball extracted is red" and the event "The second ball extracted is red" independent?

1.5.5 Odds and Bayes Formula

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